

tance, too. This is interesting not because he is a baby, but because both his arms were amputated. He now has artificial arms—and they seem to be doing well.

Sullivan's arms are a major advancement: we've had externally-powered hands and wrists and more for many years, but the problem is with the control mechanisms. These things interface with the body at the point where the natural limb was amputated; they are controlled by brain signals from a pair of muscles there. That allows only a single motion at a time; for example, to move the wrist and the elbow at the same time—try it, it's easy for us, but impossible with those prostheses. The system Sullivan has works on a different principle: researchers at the Rehabilitation Institute of Chicago discovered that when a limb is lost, the body's mechanism that controls the limb—the control signals—are accessible from nerves that can be grafted to *other* muscles. This produces signals that can be used to control prostheses in a more natural way.

Such a "muscle reinnervation procedure" is used in Sullivan's Bionic Arm. Nerves from the shoulder were transferred, in Sullivan's case, to his chest muscles. The prosthetic arm uses the signals from the chest, activated by Sullivan's thought-generated nerve impulses. As you can guess, electrodes on the surface carry the signals to the arm. Sullivan doesn't need to think "somewhere in his chest" about moving his arm; the nerves have been transferred, so when he thinks about moving his

What kind of computer signal would a nerve be able to understand?



He's actually repairing a bike! Make no mistake—prostheses today allow recipients to be near-normal

elbow, some chest muscle detects the thought, and the arm moves. It probably takes a lot of getting used to, but like we said, Jesse Sullivan seems pleased. This is semi-mind-control, because nothing is connected directly to the brain; we'll come to total mind control in a bit.

"Smarts" can be infused into anything these days, it seems. BBC reporter Stuart Hughes has a prosthetic leg. He lost his leg stepping on a landmine covering the Iraq war. Receiving a prosthetic in such a situation is routine nowadays; what's special here is that the foot—called the Proprio—is intelligent. It can detect what kind of terrain is being walked upon.

"Every type of terrain has a distinct 'signature' which the software controlling the foot is able to interpret. When the microprocessor recognises a change in terrain, it instructs a motor to automatically adjust the angle of the foot in preparation for the next step," explains Richard Hirons of Ossur, the Iceland-based prosthetics company that developed the Proprio.

The idea that we're quickly moving towards seamless integration of man and machine is getting more and more commonplace. Saeed Zahedi, visiting professor in prosthetics at the University of Surrey, is confident that we could soon create hybrid limbs that combine natural tissue with electromechanical devices. "What we're seeing at the moment is just the tip of the iceberg."

And then, there's direct mind control. A company called Cyberkinetics developed a brain-computer interface called BrainGate, which was breakthrough enough to have been in the news the world over. It is actually plugged right into the brain, and allows the paralysed patient to do things like control a computer cursor and move a robotic hand by "thought control," meaning thinking about the action does it. Matthew Nagle, stabbed in 2002 and paralysed from the neck down, was the first to try BrainGate. Of course, he still had to sit in a wheelchair all the time, but he wowed people by controlling a computer cursor and similar feats. (He actually could beat some people at Pong.)

Space doesn't permit talk of all the cool, groundbreaking stuff going on in the space of prosthetics and their interfaces and the signals that control them—see box *Links* for more. We should remember that much of this borders on "cyborgism," for lack of a better word.

We mentioned the performance artist and would-be cyborg, Stelarc, in our June 2005 issue—he quoted him as saying, "The body is neither a very efficient nor very durable structure..." He now has an extra ear in his arm and says it could be "connected to a modem and a wearable computer, and broadcast RealAudio sounds to augment the local sounds that the actual ears hear. This "extra ear" becomes a kind of Internet antenna that telematically and acoustically scales up one of the body's senses." Stelarc is not very articulate, and what follows and precedes those sentences doesn't make much sense to us. (As an aside, his Web page is, surprisingly, ghastly.) Why we're mentioning Stelarc at all is because it indicates that bionics, as a phenomenon, has excited the imagination of this new breed of humans that wants to be wedded to machines.

What's Ahead?

Plenty. If you've been with us thus far, it might just seem that whatever's been done only needs to be improved upon. Not so. Think about one of the essential problems: how to *directly interface* devices, such as implants, to the nervous system. Brand-new technology needs to be developed for this to happen. For one example, there's micro-mechanics, as we learn from *Strategic Workshop on Future Challenges in Bionics*, by Giulio Sandini of the University of Genova. Micro-mechanics would have to come in if a direct interface between nerve cells